



CSCI 340: Databases (Fall 2025)

Class time: Tuesday/Thursday, 8:00–9:50am (In-Person)

Location: Paine 212

Instructor	Babafemi Sorinolu ○ Email: babafemi.sorinolu@houghton.edu ○ Office: Paine 209B; 585.567.9259 ○ Office hours: Tuesday/Thursday, 2:30-4:30pm (Feel free to email me to schedule a virtual meeting.)
Moodle Page	https://moodle.houghton.edu/course/view.php?id=29602
Course Description	(4 credit hours) - Introduction to relational databases. Fundamentals of database and query design. Database management topics include security, integrity, and concurrency techniques. Use of relational database software (including SQL) for application projects. Topics include decision- and object-based databases. Exposure to database manager responsibilities. Prerequisite: CSCI 218 or permission.
Assigned Textbook	Raghu Ramakrishnan and Johannes Gehrke , Database Management Systems, 3rd Edition ISBN10: 0072465638 (Optional)
Required Software	Here are the general requirements: • Python and SQLite • WAMP/LAMP/MAMP Server • Programming language of your choice running SQL connection software • MongoDB

Long-Range Goals:

- 1. Understand the Relational Database Model:** One of the primary goals of this course is to understand the relational database model. Students will explore the foundational principles of relational data through both theoretical presentations and numerous practical examples.
- 2. Learn Relational Database Programming with SQL:** Students will be introduced to relational database programming using the SQL language. SQL will be applied in stand-alone environments as well as through interfaces with Python and potentially other programming languages.
- 3. Explore Semi-Structured and Unstructured Data:** The course will also cover the concepts of semi-structured and unstructured data, with a focus on XML representations and NoSQL databases.

Course Objectives: After successful completion of this course, students will be able to:

- Analyze the application of database systems to problems across a variety of disciplines.
- Understand the relational model and learn how to design relational databases using data normalization.

- Create and query databases using SQL, and improve their performance with views, indexes, and other advanced features.
- Understand the use of SQL in a server environment, and
- Design databases for unstructured data and compare how data is stored and retrieved in XML versus relational databases.

Course Time Usage: It is expected that for a 4-credit class, the well-prepared student will spend 8-12 hours on this course outside of class time.

Evaluation and Grading: The grading and grading scale will be based on the following tables:

- **Grading:** The evaluation will be based on the following criteria and weights.

Criteria	Weights (%)
Hands-on Activities	30
Homework	30
Midterm	20
Final Exam/ Project	20

- **Grading Scale**

Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	F
%	100-93	92-90	89-87	86-83	82-80	79-77	76-73	72-70	69-67	66-63	62-60	59-

All final grades will be rounded up to the nearest whole number when assigning letter grades.

Attendance: Let me know (by email to babafemi.sorinolu@houghton.edu with the subject “CSCI 340 Attendance”) if and why you will be missing a class. In general, this should be done before the class you will be missing.

Exams: There will be one midterm exam in the class and one final exam. Make-up exams will only be given in extenuating circumstances; prior arrangements must be made if possible.

Tentative dates for each exam are as follows:

- Midterm Exam: Tuesday, October 7th, 8:00–9:50am.
- Final Exam: Tuesday, December 16th, 8:00am-10:00am.

Late Submission Policy: Assignments are due at the stated deadline. Late submissions will be accepted for up to **one week with a 20% deduction**. After one week, the assignment will receive no credit unless prior arrangements have been made or there are documented extenuating circumstances. **Exams, quizzes, and in-class activities cannot be submitted late.**

Academic Honesty: Honesty is the foundation on which all intellectual endeavors rest. To use the ideas of others without acknowledging the authors of those ideas belies the nature and purpose of academic life. At Houghton, where we strive to live out Christian calling and commitment, personal integrity, including academic honesty, should be the hallmark of all our work and relationships. Students are expected to exhibit extreme care relative to personal honesty in all academic work, including in-class and out-of-class learning experiences, such as exams, quizzes, journals, papers, and research projects. Please refer to the catalog for the details of this policy ([Link](#)).

AI Policy: AI tools (e.g., ChatGPT, GitHub Copilot, Gemini) may be used in this course **as aids for learning and exploration**—for example, to review SQL syntax, generate practice queries, or clarify database concepts. However, **all submitted work must be your own**, and you are expected to fully understand any code, queries, or designs you submit. AI use must be disclosed briefly in your assignment (e.g., “Used ChatGPT to suggest alternative join syntax”). AI tools are not permitted

on exams or closed assessments. Misuse (e.g., submitting AI-generated work you cannot explain) will be treated as an academic integrity violation.

Self-Reporting of Disabilities: If you have an academic or physical disability that requires accommodations, it is up to you to self-report any such disability to the office of Academic Support and Accessibility Services in the Center for Student Success located on the 1st floor of Chamberlain. (585-567-9622) With appropriate documentation, you will be afforded the necessary accommodation. For more information about the Center for Academic Success and Advising, go to: <https://www.houghton.edu/current-students/center-for-student-success/academic-support-and-accessibility-services/>

Tentative Schedule: Below is a tentative calendar for the topics we will cover during each class period. These topics correspond to the chapters in our textbook.

Week	Class Date	Lecture Topics
1	9/2 (Tues)	The Worlds of Database Systems
	9/4 (Thurs)	The Relational Model of Data; Overview of Data Models
2	9/9 (Tues)	Data Modeling Using the Entity-Relationship (ER) Model
	9/11 (Thurs)	
3	9/16 (Tues)	Enhanced Entity-Relationship (EER) Modeling
	9/18 (Thurs)	
4	9/23 (Tues)	The Relational Data Model and Relational Database Constraints
	9/25 (Thurs)	Basic SQL
5	9/30 (Tues)	Basic SQL
	10/2 (Thurs)	
6	10/7 (Tues)	Exam #1
	10/9 (Thurs)	More SQL: Complex Queries, Triggers, Views, and Schema Modification
7	10/14 (Tues)	More SQL: Complex Queries, Triggers, Views, and Schema Modification
	10/16 (Thurs)	
8	10/21 (Tues)	October break;
	10/23 (Thurs)	The Relational Algebra and The Relational Calculus (plus QBE-Appendix C)
9	10/28 (Tues)	Relational Database Design by ER- and EERR-to-Relational Mapping
	10/30 (Thurs)	XML: Extensible Markup Language Semi-structured data Introduction to MongoDB
10	11/4 (Tues)	Transaction Management and Concurrency Control
	11/6 (Thurs)	Building Database Application
11	11/11 (Tues)	Advanced Topics in Relational Databases Security and User Authorization in SQL Recursion in SQL
	11/13 (Thurs)	Building Database Application
12	11/18 (Tues)	Business Intelligence and Data Warehouses On-Line Analytic Processing Data Cubes
	11/20 (Thurs)	Building Database Application
13	11/25 (Tues)	Overview of storage and indexing

	11/27 (Thurs)	Thanksgiving break commences;
14	12/2 (Tues)	Basics of Functional Dependencies and Normalization for Relational Databases
	12/4 (Thurs)	Database Administration and Security
15	12/9 (Tues)	Database Administration and Security
	12/11 (Thurs)	Review
16	12/16	Final Exams